

# The Smooth Manifold Structure of Lie Groups

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**[ABSTRACT]:** Sophus Lie was the founder of Lie group theory. He observed that, if a differential equation retains its form under a continuous transformation, such as a rotation or translation, this symmetry can help us simplify and solve the equation. This thesis presents reading notes and a synthesis of Lie groups as geometric objects. It first gives examples of Lie groups, then describes the geometric features that distinguish Lie groups from general manifolds, especially the way left translations extend geometric properties and quantities throughout a group. It next introduces, at an elementary level, Sophus Lie's most revolutionary idea in Lie group theory: linking Lie groups to Lie algebras so that linear tools can be used to study the intricate structure of groups. Finally,  $SU(2)$  is used to illustrate the preceding discussion.

**[Key words]:** Lie group, Lie algebra,  $SU(2)$

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# 1. Introduction and Basic Definitions: Unifying Algebraic and Geometric Objects

Since the nineteenth century, symmetry has been a central topic in mathematics and theoretical physics. While studying methods for solving differential equations, the Norwegian mathematician Sophus Lie discovered that, just as Galois used group theory to study the roots of algebraic equations, the continuous symmetries of differential equations could also be described by an algebraic structure. This profound insight ultimately gave rise to Lie group theory, elevating Lie groups from mere transformation groups to important objects in modern mathematics with the dual character of manifold topology and group-theoretic algebra. The distinctive feature of a Lie group is the seamless integration of its geometric structure and algebraic operations. As a smooth manifold, a Lie group possesses local topological properties; through its group action, it also exhibits a high degree of global homogeneity. In particular, left translations give rise to many favorable properties that general manifolds lack. They allow us to derive geometric quantities and distributions of vector fields across the entire group by studying the tangent space at the identity, namely its Lie algebra. This approach of describing a nonlinear manifold through a linear structure simplifies complex geometric problems and laid a foundation for subsequent descriptions of symmetry in physics.

## 1.1 Definition of a Lie Group

A Lie group is a mathematical object with both a geometric smooth manifold structure and an algebraic group structure. A topological manifold structure consists of a topological space  $(M, \tau)$  satisfying:

1. Local Euclidean property: for every point  $p$ , there is an open neighborhood  $U$  of  $p$  homeomorphic to an open subset of  $\mathbb{R}^n$ , for a fixed  $n$ . The manifold is then called  $n$ -dimensional.
2. C2 (the second axiom of countability): the topology has a countable base.
3. T2 (the second separation axiom): any distinct points  $p$  and  $q$  have respective neighborhoods  $U$  and  $V$  with empty intersection.

A smooth manifold  $(M, \tau, A)$  is a topological manifold  $(M, \tau)$  equipped with a smooth atlas  $A$  whose chart domains cover  $M$  and whose transition maps  $\mathbb{R}^n \rightarrow \mathbb{R}^n$  are  $\mathbb{C}^k$  (usually  $\mathbb{C}^\infty$ ). The atlas  $A$  is called a smooth structure.

A group structure consists of a set  $G$  equipped with an operation  $*$  satisfying the following four axioms:

1. Closure:  $\forall a, b \in G, a * b \in G$ .
2. Associativity:  $(a * b) * c = a * (b * c)$ .
3. Identity:  $\exists e \in G \text{ s.t. } \forall a \in G, e * a = a * e = a$ .
4. Inverses:  $\forall a \in G \exists a^{-1} \in G \text{ s.t. } a * a^{-1} = a^{-1} * a = e$ .

Two additional conditions are required for a Lie group:

1. Smooth multiplication:  $m : G * G \rightarrow G \quad (g, h) \rightarrow gh$  is a smooth map.
2. Smooth inversion:  $i : G \rightarrow G, a \rightarrow a^{-1}$  is a smooth map.

## 1.2 Examples

### 1.2.1 General Examples of Lie Groups

The definition shows that a Lie group possesses both geometric and algebraic structures and satisfies all the properties above. These conditions may seem so restrictive that examples appear difficult to find. In fact, many examples are familiar. The following are particularly evident or important examples.

Ex1  $\mathbb{R}$  is a Lie group. The real numbers  $\mathbb{R}$  form an abelian group under addition. Moreover,  $\mathbb{R}$  naturally has a one-dimensional smooth manifold structure with the identity chart  $(\mathbb{R}, \text{id})$ . Addition and negation are smooth, so  $\mathbb{R}$  is a Lie group.

Ex2  $S^1$  is a Lie group. Here  $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ , with complex multiplication as the group operation.

Ex3  $GL(V)$ , the group of linear automorphisms. We have  $GL(V) = \{A : V \rightarrow V \mid A \text{ is a linear isomorphism}\}$ . Once a basis of  $V$  is chosen, every linear map can be written explicitly as a matrix. Thus  $GL(V)$  is isomorphic to  $GL_n(K)$  and is a Lie group. Multiplication is composition of linear maps.

Ex4  $N\text{-torus} : T^n = S^1 \times S^1 \times \cdots$ . Since  $S^1$  is itself a Lie group, and a direct product of Lie groups is naturally a Lie group with componentwise multiplication,  $S^1$  being a Lie group implies that  $T^n$  is one as well.

Ex5 Clearly, every countable space with the discrete topology is a smooth manifold. Hence every countable group with the discrete topology is a Lie group.

### 1.2.2 Matrix Lie Groups and Their Topological Properties

In applications to physics and engineering, matrix Lie groups are arguably the most important and widely studied Lie groups.

(1)  $GL_n(\mathbb{R}) = \{M \in \text{Mat}_{n \times n}(\mathbb{R}) \mid \det(M) \neq 0\}$  is a Lie group. By arranging the entries of an element of  $GL_n(\mathbb{R})$  from top to bottom and left to right, we may regard it as an element of  $\mathbb{R}^{n^2}$ . Thus  $GL_n(\mathbb{R})$  inherits the subspace topology from  $\mathbb{R}^{n^2}$ . It can be viewed as  $\mathbb{R}^{n^2}$  with the hypersurface of matrices having determinant zero removed. The case of  $GL_n(\mathbb{C})$  is analogous. Below,  $K$  denotes  $\mathbb{R}$  or  $\mathbb{C}$ .

(2) Define  $SL_n(K) := \{A \in GL_n(K) \mid \det(A) = 1\}$ . This is clearly a subgroup of  $GL_n(K)$ . To prove that it is a Lie group, it suffices to show that it is a smooth submanifold, or equivalently that 1 is a regular value of the determinant function  $\det(\cdot): GL_n(K) \rightarrow K$ . Let  $A \in GL_n(K)$  have entries  $(a_j^i)$ . Write  $m_j^i$  for the  $i, j$  minor of  $A$ , namely the determinant of the  $(n-1) \times (n-1)$  matrix obtained by deleting row  $i$  and column  $j$ . Fixing  $i$ , expansion along that row gives:

$$\det(A) = (-1)^{i+1} a_1^i m_1^i + (-1)^{i+2} a_2^i m_2^i + \cdots + (-1)^{i+n} a_n^i m_n^i.$$

Observe that  $m_k^i$  contains no  $a_j^i$  for any  $k$ . Hence:

$$\frac{\partial \det A}{\partial a_j^i} = (-1)^{i+j} m_j^i.$$

In particular,  $A$  is a critical point of  $\det(A)$  (all partial derivatives vanish) if and only if all minors vanish. The preceding formula then gives  $\det(A) = 0$ , but in that case  $A \notin SL_n(K)$ . Thus the differential of  $\det(\cdot)$  is nonzero on the fiber over 1. Consequently 1 is a regular value and its fiber  $SL_n(K)$  is a smooth manifold, of dimension  $n^2 - 1$  when  $K = \mathbb{R}$  and  $2(n^2 - 1)$  when  $K = \mathbb{C}$ . Therefore  $SL_n(\mathbb{R})$  is a smooth submanifold of  $GL_n(\mathbb{R})$ .

(3)  $O(n)$  and  $SO(n)$ , the orthogonal and special orthogonal groups. Orthogonal matrices satisfy  $Q^t Q = I$ . This set is defined by  $n^2$  real variables subject to  $n(n+1)/2$  constraint equations, giving dimension  $n^2 - n(n+1)/2$ . It has two connected components:  $SO(n)$ , the determinant-1 matrices representing rotations, and  $O(n) \setminus SO(n)$ , the determinant-(-1) matrices representing a reflection followed by a rotation. We now verify that  $O(n)$  is a smooth

manifold:

$$f: GL_n(\mathbb{R}) \rightarrow S_n = \{A \in M_{\mathbb{R}}^{n \times n} \mid A^T = A\}$$

$$A \mapsto A^T A$$

To prove that  $A$  is a regular point, compute its differential:

$$df_A(X) = X^T A + A^T X$$

Given  $B \in S_n$ , we must find  $X$  such that  $df_A(X) = B$ . It suffices to solve:

$$A^T X = \frac{1}{2} B,$$

For then we also have:

$$X^T A = (A^T X)^T = \frac{1}{2} B^T = \frac{1}{2} B$$

Since  $\det(A) = \pm 1$ , we may take:

$$X = \frac{1}{2} (A^T)^{-1} B$$

This shows that  $O(n)$  is a closed submanifold of dimension:

$$n^2 - n(n+1)/2 = \frac{n(n-1)}{2}$$

It is in fact compact, since  $\text{Tr}(A^T A) = n$  is the sum of the norms of the column vectors. Thus  $O(n)$  is a submanifold of  $\mathbb{R}^{n^2}$  (in fact, of  $GL_n(\mathbb{R})$ ). Orthogonal matrices can also be defined as matrices preserving the real inner product.

Another interesting fact is that  $SO(3)$  is homeomorphic to  $RP^3$ . Let  $\bar{B}_\pi^3 = \{v \in \mathbb{R}^3 \mid \|v\| \leq \pi\}$  be the closed ball of radius  $\pi$ . Define

$$\Phi: \mathbb{R}^3 \rightarrow \mathbb{R}^9; \bar{B}_\pi^3 \rightarrow SO(3)$$

by mapping  $v$  to a rotation matrix using Rodrigues' formula:

$$\Phi(v) = I + (\sin \theta) K + (1 - \cos \theta) K^2$$

For any vector  $v$  in the ball, its length  $\theta = \|v\|$  represents the rotation angle, and the unit vector  $u = v/\|v\|$  represents the rotation axis. Here  $K$  is the cross-product matrix associated with  $u$ , with  $Kx = u \times x$ . This gives continuity; in particular, when  $v = 0$ ,  $\Phi(0) = I$ . On the

boundary  $\|\mathbf{v}\| = \pi$ , corresponding to a rotation of  $180^\circ$ , rotating by  $\pi$  about  $u$  gives exactly the same matrix as rotating by  $\pi$  about  $-u$ . Thus  $\Phi(\pi u) = \Phi(-\pi u)$  for every unit vector  $u$ . Hence  $\Phi$  is injective in the interior but identifies antipodal points on  $S^2$ . Taking the quotient by this boundary identification produces a continuous bijection from  $RP^3$  to  $SO(3)$ .

The preceding proofs proceed directly from definitions. There is also a famous result, the closed subgroup theorem [1]: if  $G$  is a Lie group and  $H$  is a topologically closed subgroup of  $G$  (that is,  $H$  is closed in the topology of  $G$  and is a subgroup under its operations), then  $H$  is a smooth submanifold of  $G$ . We may therefore consider subgroups of  $M_n(\mathbb{C})$ , viewing  $M_n(\mathbb{C})$  as  $\mathbb{C}^{n^2}$  with the topology of  $\mathbb{C}^{n^2}$ . If a subgroup  $H$  is closed in  $M_n(\mathbb{C})$ , then  $H$  is a matrix Lie group. Equivalently,  $H$  contains all its limit points: if  $A_m$  is a sequence of matrices in  $H$  and  $A_m$  converges componentwise to  $A$ , then either  $A \in H$  or  $A$  is not invertible. This criterion allows us to identify matrix Lie groups quickly.

Some common matrix Lie groups follow.

(4)  $U(n)$ , the unitary group, satisfies  $A^\dagger A = I$ , where  $A^\dagger$  denotes  $\bar{A}^T$ . Analogously,  $SU(n)$  consists of special unitary matrices with determinant 1. Another characterization is preservation of the Hermitian inner product.

(5)  $O(n; \mathbb{C})$  and  $SO(n; \mathbb{C})$ , the complex orthogonal groups, satisfy  $A \in M_n(\mathbb{C})$ ,  $A^T A = I$ . They are used less frequently and can also be defined by preservation of a complex symmetric bilinear form.

(6) The elements  $A \in GL_n(\mathbb{R})$  of  $O(n; k)$  and  $SO(n; k)$  preserve the form

$$[\cdot, \cdot]_{n,k} : [x, y]_{n,k} = \sum_{i=1}^n x_i y_i - \sum_{j=n+1}^{n+k} x_j y_j$$

If  $A(i)$  denotes column  $i^{th}$  of  $A$ , it satisfies:

1.  $[A(i), A(j)]_{n,k} = 0, j \neq l$
2.  $[A(i), A(j)]_{n,k} = 1, j \in [1, n]$
3.  $[A(i), A(j)]_{n,k} = -1, j \in [n+1, n+k]$

Let  $g = \text{diag}(1, 1, \dots, -1, \dots, -1)$ , with  $n$  initial entries equal to 1 and the remaining  $k$  entries equal to -1. Equivalently,  $A$  satisfies  $A^T g A = g$ .

(7)  $Sp(n, \mathbb{R}) \in GL_n(\mathbb{R})$ , the symplectic matrices, preserve the symplectic inner product

$$w(x, y) : \sum_{j=1}^n (x_j y_{n+j} - x_{n+j} y_j)$$

Equivalently, if  $\Omega = (0, I; -I, 0)_{2n \times 2n}$ , then  $-\Omega A^T \Omega = A^{-1}$ . Similarly, define  $Sp(n, \mathbb{C}) \in GL_n(\mathbb{C})$  by the same condition: with  $\Omega = (0, I; -I, 0)_{2n \times 2n}$ , one has  $-\Omega A^T \Omega = A^{-1}$ .

(8) Finally,  $Sp(n)$  is the compact symplectic group,  $Sp(n) = Sp(n; \mathbb{C}) \cap U(2n)$ . It preserves both the Hermitian and unitary inner products.

**Topological properties of matrix Lie groups.** The most important are compactness (equivalent to being bounded and closed, with closedness taken in  $M_n(\mathbb{C})$ ), connectedness, and simple connectedness. For example,  $O(n)$  is compact, as we now prove.

(1) Boundedness: for any orthogonal matrix  $A \in O(n)$ , the definition gives  $A^T A = I$ . Thus every column vector  $a_i$  of  $A$  is a unit vector, satisfying

$$|a_i|^2 = \sum_{j=1}^n A_{ji}^2 = 1$$

Clearly,  $A_{ji}$  satisfies  $-1 \leq A_{ji} \leq 1$ , so the set is bounded.

(2) Closedness:  $O(n)$  is the solution set of the equation

$$f(A) = A^T A - I = 0$$

The map  $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$  is continuous in the matrix entries, since it involves only multiplication and addition. The preimage of the singleton  $\{0\}$  is closed. Hence  $f^{-1}(0)$ , namely  $O(n)$ , is closed. The preimage of the singleton  $\{0\}$  remains closed; therefore  $f^{-1}(0)$ , which is  $O(n)$ , is closed. Similarly,  $SO(n), U(n)$  are compact.

For Lie groups, connectedness and path connectedness are equivalent. A Lie group is a smooth manifold, hence locally Euclidean and therefore locally path connected. By a standard topological result, a connected, locally path-connected space is path connected.

For example,  $GL_n(\mathbb{R})$  is not connected. Its two connected components are  $GL_n^+(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) \mid \det(A) > 0\}$  (containing the identity matrix) and  $GL_n^-(\mathbb{R}) = \{A \in GL_n(\mathbb{R}) \mid \det(A) < 0\}$ . In contrast,  $GL_n(\mathbb{C})$  is connected.

Matrix Lie groups are widely studied by mathematicians and physicists, with key roles in rigid-body motion, gauge field theory, and quantum mechanics. This brief account will be revisited in the final chapter.

## 2. Special Manifold Properties of Lie Groups

When a Lie group is viewed as a manifold, left and right translations provide the most elegant and important manifestations of its group structure. By left or right multiplication, they extend geometric quantities and properties at the identity throughout the group, yielding favorable properties absent from general manifolds. Left translation is more than a simple group operation: it connects local algebra with global geometry. Without left translations, a Lie group would be merely an ordinary differentiable manifold; with them, it becomes a homogeneous space under its action on itself. Their central value is that a quantity or property defined at the identity  $e$  can be replicated consistently throughout the manifold.

### Special Homomorphisms: Left and Right Translations

Set

$$\begin{aligned} L_g : G &\rightarrow G, & L_g(h) &= gh \\ R_g : G &\rightarrow G, & R_g(h) &= hg \end{aligned}$$

$L_g$  is an isomorphism because it is a homomorphism induced from multiplication  $G * G \rightarrow G$  by fixing the first factor  $g$ , giving a map  $G \rightarrow G$ . Since multiplication is smooth,  $L_g$  is smooth, and its inverse  $x \mapsto g^{-1}x$  is also smooth. Likewise,  $R_g$  is an isomorphism. In particular, they are diffeomorphisms, whose differentials are linear isomorphisms between tangent spaces. These maps play a crucial role in the geometric quantities and properties discussed below.

### 2.1 Lie Groups Always Admit Left-Invariant Vector Fields

On a manifold  $M$ , a tangent vector is a derivation, and a smooth vector field  $\mathfrak{X}(M)$  is a global section  $\Gamma(TM)$ . Smoothness means that the section  $s : M \rightarrow TM$  is smooth. By the topology and smooth structure of  $TM$  as a vector bundle, a vector field  $X$  on an open set  $U \subset M$  is smooth if and only if its component functions in every (equivalently, some) local coordinate chart are smooth. In particular, smooth fields on  $U$  and  $V$  that agree on their overlap define a smooth field on  $U \cup V$ . Conversely, smoothness can be checked by expressing a field in one or more local coordinate systems as  $\sum X^i \frac{\partial}{\partial x^i}$  and verifying that its coefficients  $X^i$  are smooth functions.

Not every manifold admits a globally nonzero smooth vector field (the Poincaré-Hopf theorem). For example, every odd-dimensional sphere admits such a field, whereas no even-dimensional sphere does (the hairy ball theorem).

### 2.1.1 Left-Invariant Vector Fields on a Lie Group

A (left-)invariant vector field  $X$  satisfies  $(L_g)_*X = X$ ; explicitly, it is a field satisfying

$$d(L_g)_{g_0}(X_{g_0}) = X_{g_0g}, \forall g, g_0 \in G$$

**Proposition:** Lie groups always admit left-invariant vector fields.

**Proof:** Take any vector  $X_e$  in the tangent space  $T_eG$  at the identity. Fix  $g$  and consider the differential of left translation

$$d(L_g)_e : T_eG \rightarrow T_gG$$

$$X_e \longmapsto d(L_g)_e(X_e)$$

Allowing  $g$  to range over  $G$  produces a global vector field through the differentials of left translations. It is indeed left-invariant. For arbitrary  $g_0, g_1 \in G$ , consider

$$d(L_{g_0g_1^{-1}})_{g_1} : T_{g_1}G \rightarrow T_{g_0}G$$

$$X_{g_1} \longmapsto d(L_{g_0g_1})_{g_1}X_{g_1}(\star)$$

We must show that  $(\star)$  equals  $X_{g_1g_0}$ , namely  $d(L_{g_0})_eX_e$ . By construction,

$$X_{g_1} = d(L_{g_1})_eX_e$$

so

$$(\star) = d(L_{g_0g_1^{-1}})_{g_1} \circ d(L_{g_1})_eX_e$$

We have

$$L_{g_0} = L_{g_0g_1} \circ L_{g_1}$$

differentiating gives

$$d(L_{g_0g_1^{-1}})_{g_1} \circ d(L_{g_1})_e = d(L_{g_0})_e$$

Thus  $(\star)$  equals  $X_{g_1g_0}$ .

We have now constructed a left-invariant field without yet establishing its regularity. In fact, this produces all left-invariant vector fields, because each is obtained by distributing its value at the identity through the differentials of left translations (set  $g_1 = e$ ). We denote the collection of all left-invariant fields by  $\text{Lie}(G)$ .

### 2.1.2 Left-Invariant Vector Fields on a Lie Group Are Nowhere Zero and Smooth

**Proposition:** Left-invariant vector fields on a Lie group are nowhere zero and smooth.

**Proof:** Let  $X \in \text{Lie}(G)$ . To verify smoothness, it suffices to show that, for every  $f \in C^\infty(G)$ ,  $X(f)$  is a smooth function.

Choose a smooth curve  $\gamma(t) \in G$  satisfying

$$\gamma(0) = e, \quad \gamma'(0) = X_e.$$

Then

$$(Xf)(e) = \left. \frac{d}{dt} \right|_{t=0} f(\gamma(t)).$$

On the other hand, since  $X$  is left-invariant, the map  $t \mapsto g\gamma(t)$  is a smooth curve through  $g$ , whose derivative at  $t = 0$  is  $X_g$ . Therefore,

$$(Xf)(g) = \left. \frac{d}{dt} \right|_{t=0} f(g\gamma(t)).$$

Since  $g\gamma(t)$  depends smoothly on both  $g$  and  $t$ ,  $(Xf)(g)$  is a smooth function of  $g$ . Hence  $X$  is smooth. This gives a favorable setting for studying flows and exponential maps.

The absence of a smooth vector field can also quickly rule out a Lie group structure. For example,  $S^1$  is a Lie group whereas  $S^2$  is not, because  $S^2$  has no nonzero smooth vector field.

## 2.2 The Tangent Bundle of a Lie Group Is Trivial

By the definition of a vector bundle, a rank- $n$  bundle is trivial if and only if it admits  $n$  global sections that are linearly independent everywhere. Since  $L_g$  is a diffeomorphism and  $dL_g$  is a linear isomorphism, choose  $n$  linearly independent vectors at the identity and push them forward over the group. They remain linearly independent at every point, yielding the required global sections.

## 2.3 Integral Curves on a Lie Group Are Complete and Their Flows Are Global

**Theorem:** Let  $G$  be a Lie group and  $X$  a left-invariant vector field on  $G$ . For every initial point  $g \in G$ , the integral curve  $\gamma(t)$  through it is complete: its domain is  $\mathbb{R}$ .

**Proof:** By local existence and uniqueness for ordinary differential equations, there exists  $\epsilon > 0$  such that the integral curve  $\gamma_e(t)$  through the identity  $e \in G$ , passing through  $e$  with tangent  $X_e$ , is defined on  $(-\epsilon, \epsilon)$  and satisfies

$$\gamma_e(0) = e$$

and

$$\dot{\gamma}_e(t) = X_{\gamma_e(t)}$$

Since  $X$  is left-invariant, for any  $a \in G$  the differential of  $L_a$  preserves  $X$ :

$$(dL_a)X = X \circ L_a$$

Thus, whenever  $\gamma(t)$  is an integral curve, its left translate

$$L_a(\gamma(t)) = a \cdot \gamma(t)$$

is also an integral curve of  $X$ .

Suppose the maximal interval of existence of  $\gamma_e(t)$  is

$$(t_{min}, t_{max})$$

If  $t_{max} < \infty$ , choose a time  $t_0$  close enough to  $t_{max}$  that  $t_{max} - t_0 < \epsilon/2$ . Set  $g_0 = \gamma_e(t_0)$ .

By left invariance, the curve

$$\sigma(t) = L_{g_0}(\gamma_e(t))$$

is also an integral curve and satisfies  $\sigma(0) = g_0$ . Since  $\gamma_e(t)$  is defined on an interval of length  $\epsilon$  near the identity, left translation gives an integral curve starting at  $g_0$  that is defined on at least  $(-\epsilon, \epsilon)$ .

Splice the original curve  $\gamma_e(t)$  to the translated curve by defining  $\tilde{\gamma}(t)$  to equal  $\gamma_e(t)$  when  $t \leq t_0$  and  $\sigma(t - t_0)$  when  $t > t_0$ . Because  $t_0 + \epsilon > t_{max}$ , this extends the solution beyond  $t_{max}$ , contradicting the maximality of  $t_{max}$ . Therefore  $t_{max} = \infty$ . Similarly,  $t_{min} = -\infty$ . Hence integral curves of left-invariant vector fields are complete.

## 2.4 Lie Groups Admit Nowhere-Vanishing Volume Forms

**Theorem:** A Lie group admits a nowhere-vanishing volume form.

**Proof:** Let  $G$  be a Lie group. At the identity  $e$ , choose a basis  $\{v_1, \dots, v_n\}$  of the tangent

space  $T_e G$  and define a nonzero  $n$ -form

$$\omega_e = v_1^* \wedge \cdots \wedge v_n^*$$

Extend it globally by left translation: using  $L_g$ , define a global  $n$ -form  $\omega$  by setting, for every  $g \in G$ ,

$$\omega_g = (L_{g^{-1}})^* \omega_e$$

This pulls the form at the identity back to  $g$ . It is nowhere zero: since  $L_g$  is a diffeomorphism, its pullback preserves nonvanishing. By definition, applying  $\omega_g$  at  $g$  to tangent vectors  $X_1, \dots, X_n \in T_g G$  gives

$$\omega_g(X_1, \dots, X_n) = \omega_e(d(L_{g^{-1}})_g(X_1), \dots, d(L_{g^{-1}})_g(X_n))$$

Therefore  $\omega$  is a smooth, nowhere-vanishing  $n$ -form on  $G$ .

This left-invariant volume form makes analysis on the Lie group possible. It defines the Haar measure

$$\mu(E) = \int_E |\omega|$$

Extending a top-degree form  $\omega$  by  $\omega_g = (L_{g^{-1}})^* \omega_e$  gives left invariance, ensuring translation invariance of integration:  $\int_G f(gx) d\mu(x) = \int_G f(x) d\mu(x)$ . Without left translations, we could not define a consistent average, convolution, or Fourier transform on the group.

### 2.4.1 Compact Lie Groups Admit Bi-Invariant Volume Forms and Haar Measures

**Theorem:** A Lie group admits a bi-invariant volume form and a bi-invariant Haar measure.

**Proof:** Let  $\omega$  be a nonzero left-invariant  $n$ -form on  $G$ , so that for all  $h \in G$ ,  $L_h^* \omega = \omega$ . For fixed  $g \in G$ , consider the pullback  $R_g^* \omega$  of  $\omega$  under right translation  $R_g : G \rightarrow G$ . To check that  $R_g^* \omega$  remains left-invariant, take any  $h \in G$ :

$$L_h^*(R_g^* \omega) = (R_g \circ L_h)^* \omega$$

Left and right translations commute, since  $R_g(L_h(x)) = hxg = L_h(R_g(x))$ , so

$$(R_g \circ L_h)^* \omega = (L_h \circ R_g)^* \omega = R_g^*(L_h^* \omega)$$

Because  $\omega$  is left-invariant ( $L_h^*\omega = \omega$ ), this becomes

$$R_g^*(L_h^*\omega) = R_g^*\omega$$

Thus  $R_g^*\omega$  is also a left-invariant  $n$ -form. Step 2: introduce a proportionality factor (the modular function). On an  $n$ -dimensional manifold, the space of top-degree forms at each point is 1-dimensional. A left-invariant  $n$ -form is determined by its value at  $e$ , so the space of left-invariant  $n$ -forms on  $G$  is 1-dimensional. Hence  $\omega$  and  $R_g^*\omega$  satisfy

$$R_g^*\omega = \lambda_g \omega$$

where  $\lambda_g$  is a real constant depending only on  $g$ . Next use compactness to normalize: since  $G$  is compact and  $\omega$  is a smooth, nowhere-vanishing volume form with a chosen orientation, its integral is finite and nonzero:

$$V = \int_G \omega \quad (0 < |V| < \infty)$$

Rescale  $\omega$  so that  $V = 1$ . Step 4: prove bi-invariance, namely  $\lambda_g = 1$ . Invariance of integration under diffeomorphisms, or change of variables over the whole manifold, gives

$$\int_G R_g^*\omega = \int_G \omega = 1$$

Substituting  $R_g^*\omega = \lambda_g \omega$  yields

$$\int_G \lambda_g \omega = \lambda_g \int_G \omega = \lambda_g \cdot 1 = 1$$

and therefore

$$\lambda_g = 1 \quad (\forall g \in G)$$

Thus, for every  $g \in G$ ,  $R_g^*\omega = \omega$ .

## 2.5 Lie Groups Are Always Orientable

A smooth manifold with a trivial tangent bundle is necessarily orientable. Likewise, the existence of a nowhere-vanishing volume form implies orientability.

## 2.6 Lie Group Homomorphisms Always Have Constant Rank

**Proposition:** A Lie group homomorphism has constant rank: the rank of  $dF$  is the same at every point.

**Proof:** Fix  $g_0$ . For  $\forall g \in M$ ,

$$F(L_{g_0}(g)) = F(g_0g) = F(g_0)F(g) = L_{F(g_0)}(F(g))$$

Thus

$$F \circ L_{g_0} = L_{F(g_0)} \circ F$$

Differentiating gives

$$dF_{g_0} \circ (dL_{g_0})_e = (dL_{F(g_0)})_{\hat{e}} \circ dF_e$$

Since left translations are isomorphisms, their differentials are isomorphisms, so

$$\text{rank}(dF_{g_0}) = \text{rank}(dF_e)$$

For  $g_0$ , the map  $L_{g_0}$  is an isomorphism and hence a diffeomorphism, whose differential is an isomorphism between tangent spaces. Since  $g_0$  is arbitrary, the conclusion follows. Constant rank ensures that the constant-rank theorem and level-set theorem apply.

## 2.7 Lie Group Isomorphisms: A Bijective Smooth Homomorphism Has a Smooth Inverse

$F$  is a smooth group homomorphism:  $F$  is smooth and satisfies  $F(m_1 * m_2) = F(m_1) * F(m_2)$ .  $F$  is a Lie group isomorphism if it is a bijective Lie group homomorphism. Its inverse is then automatically a smooth homomorphism; equivalently,  $F$  is both a homomorphism and a diffeomorphism. Two contrasting facts about manifolds and groups illustrate the distinctive rigidity of algebraic structure.

For groups, the inverse of a bijective homomorphism automatically preserves the structure and is again a homomorphism.

For manifolds this is not true: a smooth bijection  $F : M \rightarrow N$  need not have a smooth inverse.

A classical example is the map

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$$

Here  $f$  is a bijection from the real numbers to themselves, and  $f$  is smooth, with derivatives of every order. However, the inverse  $f^{-1}(y) = y^{1/3}$  is not differentiable at  $y = 0$  and therefore is not smooth. Its derivative at 0 does not exist. Thus  $f$  is not a diffeomorphism of  $\mathbb{R}$ , even though it is a smooth bijection.

The additional rigidity of Lie group structure changes this. We shall prove that if  $F$  is a bijective Lie group homomorphism, then  $F^{-1}$  is automatically smooth.

**Proposition:** If  $F$  is a bijective Lie group homomorphism, then  $F^{-1}$  is automatically smooth.

**Proof:** Let  $\phi : G \rightarrow H$  be a bijective smooth homomorphism of Lie groups. Consider its differential at the identity:

$$d\phi_e : T_e G \rightarrow T_e H$$

Since  $\phi$  is a group homomorphism (see Chapter 3), its associated Lie algebra map  $d\phi_e$  is a homomorphism between linear spaces. We show that  $d\phi_e$  is bijective.

Injectivity: for a Lie group homomorphism  $F$ ,

$$\ker(dF_e) = T_e(\ker F)$$

In other words, its Lie algebra kernel is the tangent space at the identity to its group kernel. If  $\phi$  is bijective, its kernel  $\ker(\phi) = \{e\}$  is discrete, so the kernel of  $d\phi_e$  is zero.

Surjectivity: suppose  $d\phi_e$  is not surjective. Since a homomorphism has constant rank,  $d\phi_e$  fails to have full rank throughout a neighborhood  $U$  of the identity. The constant-rank theorem gives coordinates  $\{x^i\}$  on  $U$  and  $\{y^j\}$  on a neighborhood  $V$  of the identity in  $H$  such that

$$\phi|_U(x_1, x_2, \dots, x_m) = (x_1, x_2, \dots, x_r, 0, \dots, 0)$$

Clearly,  $\phi$  is no longer bijective on  $U$ , a contradiction.

Thus  $d\phi_e$  is a linear isomorphism. Since  $d\phi_e$  is invertible, the inverse function theorem gives a neighborhood  $U$  of the identity  $e$  on which  $\phi|_U$  is a local diffeomorphism. Left translation carries this conclusion to every point, so  $\phi$  is everywhere a local diffeomorphism. Together with bijectivity, this makes it a global diffeomorphism, whose inverse is smooth.

## 2.8 Left-Invariant Metrics on Lie Groups

On a general Riemannian manifold, a metric is a  $(0,2)$ -tensor specified pointwise subject to smoothness. On a Lie group, we can choose an inner product at the identity and extend it throughout the group by left translation to obtain a well-defined Riemannian metric. This

provides a symmetric setting for discussing distance, the Levi-Civita connection, and the Riemann curvature tensor. In fact, any tensor of type (r,s) at the identity can be extended over the entire group by left translation.

### 2.8.1 Left Translations Are Isometries for a Left-Invariant Metric

A metric  $\langle \cdot, \cdot \rangle$  on a Lie group  $G$  is called left-invariant (respectively, right-invariant) if and only if

$$\langle u, v \rangle_b = \langle (dL_a)_b u, (dL_a)_b v \rangle_{ab} \quad (\text{respectively, } \langle u, v \rangle_b = \langle (dR_a)_b u, (dR_a)_b v \rangle_{ba}),$$

for all  $a, b \in G$  and  $u, v \in T_b G$ . Thus  $L_g$  is an isometry. By left invariance, any tangent vector  $v \in T_p G$  retains its length under left translation:

$$\|(dL_g)_p v\|_{gp} = \sqrt{\langle (dL_g)_p v, (dL_g)_p v \rangle_{gp}} = \sqrt{\langle v, v \rangle_p} = \|v\|_p$$

The map  $L_g$  also preserves curve length. If  $\gamma(t)$  joins  $p$  to  $q$ , its length  $L(\gamma)$  is the integral of its speed:

$$L(\gamma) = \int_a^b \|\dot{\gamma}(t)\|_{\gamma(t)} dt$$

Now consider the left-translated curve  $\sigma(t) = L_g(\gamma(t)) = g \cdot \gamma(t)$ . By the chain rule its tangent (velocity) is  $\dot{\sigma}(t) = (dL_g)_{\gamma(t)} \dot{\gamma}(t)$ . Since speed is preserved, as shown above,

$$L(\sigma) = \int_a^b \|\dot{\sigma}(t)\|_{g\gamma(t)} dt = \int_a^b \|\dot{\gamma}(t)\|_{\gamma(t)} dt = L(\gamma)$$

Hence  $L_g$  preserves the length of every curve.

A Riemannian metric that is both left- and right-invariant is called a *bi-invariant metric*. Chapter 4 gives a concrete example on a matrix Lie group. It is easy to verify that every Lie group admits a left-invariant (or right-invariant) metric, although it need not admit a bi-invariant one.

Left-invariant metrics correspond one-to-one to inner products on the Lie algebra. Likewise, bi-invariant metrics correspond one-to-one to Ad-invariant inner products on the Lie algebra.

### 2.8.2 A Lie Group with a Left-Invariant Metric Is Complete

**Proposition:** Let  $G$  be a Lie group with a left-invariant Riemannian metric  $d$ . Then  $(G, d)$  is a complete metric space.

**Proof:** The metric induces the distance

$$d(p, q) = \inf \left\{ \int_a^b \sqrt{g(\dot{\gamma}(t), \dot{\gamma}(t))} dt \right\}$$

Its induced topology agrees with the manifold topology. Let  $\{g_n\}$  be a Cauchy sequence in  $G$ . As a manifold,  $G$  is locally compact, so there is a closed ball

$$V = \{x \in G \mid d(e, x) \leq \epsilon\}$$

of radius  $\epsilon > 0$  about the identity  $e$ . Choose  $\epsilon$  small enough that the ball lies inside a Euclidean neighborhood  $U$ . It is closed and bounded in  $U$ , hence compact. For the given  $\epsilon$ , the Cauchy condition gives a positive integer  $N$  such that  $d(g_n, g_m) < \epsilon$  whenever  $n, m \geq N$ . In particular, for every  $n \geq N$ , the point  $g_n$  lies in the closed ball

$$B(g_N, \epsilon)$$

centered at  $g_N$  with radius  $\epsilon$ . By left invariance,  $L_{g_N} : G \rightarrow G$  is an isometry carrying the compact set  $V$  at the identity to the ball at  $g_N$ :

$$B(g_N, \epsilon) = L_{g_N}(V)$$

Thus  $B(g_N, \epsilon)$  is compact. Every sequence in a compact metric space has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges in the original space. Its limit  $g^*$  belongs to  $B(g_N, \epsilon) \subset G$ , so  $\{g_n\}$  converges in  $G$ . This proves the proposition. If we further assume that the Lie group  $G$  is simply connected, the Hopf-Rinow theorem gives the following consequences for a left-invariant metric:

1.  $G$  is geodesically complete: all geodesics extend to  $\mathbb{R}$ .
2. Every closed bounded subset is compact (the Heine-Borel property).
3. At any point  $p$ , the Riemannian exponential map is defined on the entire tangent space  $T_p M$ .
4. Any two points can be joined by a minimizing geodesic.

### 2.8.3 Every Compact Lie Group Admits a Bi-Invariant Metric

**Proposition:** Every compact Lie group admits a bi-invariant metric.

**Proof:** Begin with any inner product  $\langle \cdot, \cdot \rangle_0$  on  $\mathfrak{g}$ . Since  $G$  is compact, it has a bi-invariant Haar measure (equivalently, a bi-invariant volume form), denoted  $\omega$ . Average over the group and define

$$\langle v, w \rangle_e := \int_G \langle \text{Ad}_g v, \text{Ad}_g w \rangle_0 \omega$$

This gives a new inner product on  $\mathfrak{g}$ , which is Ad-invariant. Indeed, for any  $h \in G$ ,

$$\langle \text{Ad}_h v, \text{Ad}_h w \rangle_e = \int_G \langle \text{Ad}_g \text{Ad}_h v, \text{Ad}_g \text{Ad}_h w \rangle_0 \omega$$

Using

$$\text{Ad}_g \text{Ad}_h = \text{Ad}_{gh}$$

and right invariance of Haar measure gives

$$\int_G \langle \text{Ad}_{gh} v, \text{Ad}_{gh} w \rangle_0 \omega = \int_G \langle \text{Ad}_g v, \text{Ad}_g w \rangle_0 \omega = \langle v, w \rangle_e$$

Thus  $\langle \cdot, \cdot \rangle_e$  is Ad-invariant, and the associated left-invariant metric is bi-invariant.

### 2.8.4 Lie Groups with Bi-Invariant Metrics Have Nonnegative Sectional Curvature

**Proposition:** The sectional curvature of a Lie group with a bi-invariant metric is always nonnegative.

**Proof:** We first establish several facts. Assume that the metric is bi-invariant and  $X, Y, Z$  are left-invariant vector fields.

1.  $\langle [X, Y], Z \rangle = \langle X, [Y, Z] \rangle$ .
2.  $\nabla_X Y = \frac{1}{2}[X, Y]$ .
3.  $\text{Exp}_e(tv) = \exp(tv), \forall v \in \mathfrak{g} = T_e G$ . In other words, the Lie group exponential map agrees with the Riemannian exponential map.

Since the metric is bi-invariant and  $Y, Z$  are left-invariant, the function

$$g \mapsto \langle Y_g, Z_g \rangle_g = \langle dL_g Y_e, dL_g Z_e \rangle_e$$

is constant on  $G$ . Evaluate this constant function along the one-parameter subgroup

$$t \mapsto g(t) = \text{Exp}(tX)$$

Using right invariance to translate back gives

$$\begin{aligned} \langle Y_{g(t)}, Z_{g(t)} \rangle_{g(t)} &= \langle dR_{g^{-1}(t)} Y_{g(t)}, dR_{g^{-1}(t)} Z_{g(t)} \rangle_e \\ &= \langle dC_{g(t)} Y_e, dC_{g(t)} Z_e \rangle_e \\ &= \langle \text{Ad}_{g(t)} Y_e, \text{Ad}_{g(t)} Z_e \rangle_e \end{aligned}$$

Differentiating at  $t = 0$  and applying the product rule yields

$$\begin{aligned} 0 &= \left\langle \frac{d}{dt} \Big|_{t=0} \text{Ad}_{\text{Exp}(tX)} Y_e, \text{Ad}_{\text{Exp}(tX)} Z_e \right\rangle_e + \left\langle \text{Ad}_{\text{Exp}(tX)} Y_e, \frac{d}{dt} \Big|_{t=0} \text{Ad}_{\text{Exp}(tX)} Z_e \right\rangle_e \\ &= \langle \text{ad}_X(Y), Z \rangle + \langle Y, \text{ad}_X(Z) \rangle \end{aligned}$$

But  $\text{ad}_X(Y) = [X, Y]$  is simply the Lie bracket, so

$$\langle [X, Y], Z \rangle = -\langle Y, [X, Z] \rangle = \langle X, [Y, Z] \rangle,$$

This proves (1). For (2), apply the Koszul formula

$$2\langle \nabla_X Y, Z \rangle = X\langle Y, Z \rangle + Y\langle Z, X \rangle - Z\langle X, Y \rangle + \langle [X, Y], Z \rangle - \langle [Y, Z], X \rangle - \langle [X, Z], Y \rangle.$$

Since  $\langle X, Y \rangle$ ,  $\langle Y, Z \rangle$  and  $\langle Z, X \rangle$  are constant, the three derivative terms vanish, giving

$$2\langle \nabla_X Y, Z \rangle = \langle [X, Y], Z \rangle - \langle [Y, Z], X \rangle - \langle [X, Z], Y \rangle.$$

Using (1), this simplifies to

$$2\langle \nabla_X Y, Z \rangle = \langle [X, Y], Z \rangle.$$

As it holds for every left-invariant field  $Z$ , we obtain

$$\nabla_X Y = \frac{1}{2}[X, Y].$$

For (3), let  $v \in \mathfrak{g}$  and let  $X$  be the corresponding left-invariant field with  $X_e = v$ . The integral curve of  $X$  through the identity  $e$  is

$$\gamma(t) = \text{Exp}(tv).$$

We have

$$\nabla_{\dot{\gamma}}\dot{\gamma} = \nabla_X X \stackrel{(2)}{=} \frac{1}{2}[X, X] = 0.$$

so  $\gamma$  is the geodesic with initial velocity  $v$ . Therefore

$$\text{Exp}(tv) = \gamma(t) = \exp(tv).$$

Now consider curvature. We have

$$\mathcal{R}(X, Y)Z = -\frac{1}{4}[[X, Y], Z]$$

and

$$Rm(X, Y, Z, W) = \frac{1}{4}\langle [X, Y], [Z, W] \rangle.$$

Proof: using

$$\nabla_X Y = \frac{1}{2}[X, Y],$$

compute

$$\mathcal{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Thus

$$\mathcal{R}(X, Y)Z = \frac{1}{4}[X, [Y, Z]] - \frac{1}{4}[Y, [X, Z]] - \frac{1}{2}[[X, Y], Z].$$

The Jacobi identity gives

$$[X, [Y, Z]] - [Y, [X, Z]] = [[X, Y], Z].$$

and therefore

$$\mathcal{R}(X, Y)Z = \frac{1}{4}[[X, Y], Z] - \frac{1}{2}[[X, Y], Z] = -\frac{1}{4}[[X, Y], Z].$$

Applying (1) again gives the sectional curvature

$$K(X, Y) = -\frac{1}{4}(-\|[X, Y]\|^2) = \frac{1}{4}\|[X, Y]\|^2$$

### 3. Basic Correspondences between Lie Groups and Lie Algebras

Studying Lie groups requires studying Lie algebras. In many settings, a Lie algebra indirectly reflects the properties of its Lie group. Their relationship is inseparable: a Lie group is a curved manifold with a group structure, hence a nonlinear object, while its Lie algebra is the tangent space at the identity, a linear object. Studying the Lie algebra often reveals information about the group. The central idea is to sacrifice very little information in return for substantial computational convenience.

#### 3.1 Abstract Lie Algebras and the Lie Algebra of a Lie Group

##### Abstract Lie Algebras

Algebraically, a Lie algebra is a real or complex vector space equipped with a Lie bracket  $[\cdot, \cdot]$  satisfying:

1. Bilinearity:  $[c_1x + c_2y, z] = c_1[x, z] + c_2[y, z]$ .
2. Antisymmetry:  $[x, y] = -[y, x]$ .
3. Jacobi Identity  $[x, [y, z]] + [y, [x, z]] + [z, [x, y]] = 0$

For example,  $R^3$  with the cross product  $[X, Y] = X \times Y$  is a Lie algebra.

##### The Lie Algebra of a Lie Group

The Lie algebra of  $G$  is defined as the collection of all its left-invariant vector fields. In general, the space  $\mathcal{X}(G)$  of all smooth fields is a module over the ring  $C^\infty(G)$ . By contrast, left-invariant fields form a vector space  $\text{Lie}(G)$ , with scalar multiplication by constants. One can verify the eight vector space axioms.

##### 3.1.1 Two Descriptions of a Lie Algebra

Let  $\mathfrak{g}$  be the tangent space at the identity. Define the map

$$\phi : \text{Lie}(G) \rightarrow \mathfrak{g}; X \mapsto X_e$$

By the construction of left-invariant fields, this is an isomorphism, so  $\text{Lie}(G)$  can be identified with  $\mathfrak{g}$ . Vector fields already possess a Lie bracket

$$[X, Y] = XY - YX$$

where the right side is defined by

$$C^\infty(M) \rightarrow C^\infty(M) ; f \mapsto X(Y(f)) - Y(X(f))$$

If  $f$  is a diffeomorphism, its pushforward  $f_*$  preserves Lie brackets:

$$f_*[X, Y] = [f_*X, f_*Y]$$

When  $f = L_g$  and  $X, Y$  are left-invariant, this also equals  $[X, Y]$ . Thus  $X, Y \in \text{Lie}(G), [X, Y] \in \text{Lie}(G)$ . The bracket therefore makes  $\text{Lie}(G)$  a Lie algebra and likewise makes  $\mathfrak{g}$  a Lie algebra, with bracket defined by

$$[X_e, Y_e] = [X, Y]|_e$$

## 3.2 Bridging Lie Groups and Lie Algebras: The Exponential Map and the Adjoint Representation

One-parameter subgroups and exponential maps give a map from the Lie algebra to the Lie group, connecting the two. Its favorable properties allow us to relate the behavior of the Lie algebra to that of the group. Likewise, the adjoint representation is crucial because it establishes a natural bridge between a Lie group, a curved geometric object, and its Lie algebra, a linear algebraic object.

Without the adjoint representation, Lie groups and Lie algebras would resemble separate subjects. With it, linear algebra can be used to analyze intricate geometric symmetries and to discuss deeper topics such as the Killing form, semisimplicity, and classification.

The exponential map is defined using one-parameter subgroups.

### 3.2.1 One-Parameter Subgroups

Let  $G$  be a Lie group. A smooth group homomorphism  $\phi : \mathbb{R} \rightarrow G$  from the additive group  $(\mathbb{R}, +)$  to  $G$  is called a one-parameter subgroup of  $G$ . The map  $\phi$  must satisfy:

1. Identity correspondence:  $\phi(0) = e$ ; zero corresponds to the identity of the group.
2. Exponential law: for any  $s, t \in \mathbb{R}$ ,  $\phi(s + t) = \phi(s) \cdot \phi(t)$ .

Here “subgroup” refers to the image  $\phi(\mathbb{R})$ , which is closed under the group operation by the homomorphism property.

**Theorem:** One-parameter subgroups coincide with complete integral curves through the identity.

**Proof:** In a Lie group, integral curves of left-invariant vector fields through the identity  $e$  are precisely one-parameter subgroups. As before, existence and uniqueness for ODEs give a unique integral curve  $\alpha(t)$  starting at  $e$  and satisfying

$$\alpha(0) = e, \alpha'(t) = \tilde{X}(\alpha(t)) = dL_{\alpha(t)}(X)$$

Since  $\tilde{X}$  is left-invariant, left-translating the entire curve  $\alpha(s)$  by  $g$  produces another integral curve  $g\alpha(s)$  of the same field, now starting at  $g$ . Indeed, write  $g\alpha(s)$  as  $L_g \circ \alpha(s)$  and differentiate in  $s$  to obtain

$$\frac{d(L_g \circ \alpha(s))}{ds} = (dL_g)_{\alpha(s)} \alpha'(s) = (L_g)_* \alpha'(s)$$

which satisfies left invariance.

We now prove the one-parameter subgroup property. Fix  $t$ .

Curve A:  $\gamma_1(s) = \alpha(t + s)$  is the original curve starting at time  $t$ .

Curve B:  $\gamma_2(s) = \alpha(t)\alpha(s)$  is the entire original curve left-translated by  $\alpha(t)$ . They satisfy:

1. The same initial point: when  $s = 0$ ,  $\gamma_1(0) = \alpha(t)$  and  $\gamma_2(0) = \alpha(t)e = \alpha(t)$ .
2. The same velocity condition: at every time  $s$ , each curve’s tangent equals the value of the vector field at that point. By uniqueness of ODE solutions, the curves coincide.

Thus

$$\alpha(t + s) = \alpha(t)\alpha(s)$$

The integral curve  $\alpha(t)$  is therefore a group homomorphism and a one-parameter subgroup. Conversely, a one-parameter subgroup satisfies  $\alpha(t + s) = \alpha(t)\alpha(s)$ . Differentiating both sides in  $s$  gives

$$X(\alpha(t + s)) = (dL_{\alpha(t)})_{\alpha(s)} \alpha'(s)$$

Setting  $s=0$  yields

$$X(\alpha(t)) = (dL_{\alpha(t)})_e X_e$$

the integral curve condition. Hence integral curves through the identity are precisely one-parameter subgroups.

### 3.2.2 The Lie Group Exponential Map

The Riemannian exponential map is defined through geodesics and relates a subset of the tangent bundle to a Riemannian manifold. Similarly, the Lie group exponential map  $\exp : \mathfrak{g} \rightarrow G$  is defined through integral curves and relates the Lie algebra to the group. It is a central bridge in Lie theory: more than a formula, it converts infinitesimal linear information into global geometric structure.

For any vector  $X \in \mathfrak{g}$  in the Lie algebra, define

$$\exp(X) = \phi_X(1)$$

where  $\phi_X$  is the unique one-parameter subgroup with initial velocity  $X$ . Properties of ODE solutions give

$$\exp(tX) = \phi_{tX}(1) = \phi_X(t)$$

**Theorem:** The exponential map is a local diffeomorphism.

**Proof:** Viewing  $\mathfrak{g}$  as a manifold gives

$$d(\exp)_0 : \mathfrak{g} \rightarrow \mathfrak{g}$$

$$d(\exp)_0(X) = \left. \frac{d}{dt} \right|_{t=0} \phi_X(t) = X$$

Thus  $d(\exp)_0 = I$ , so  $\exp$  is a local diffeomorphism near the origin. Left translation likewise gives a local diffeomorphism between a neighborhood of any  $g$  in the group and a neighborhood in the Lie algebra.

The local inverse of the exponential map is called the logarithm map. In the following sections we shall see how the Lie algebra describes local group multiplication.

The exponential map need not be surjective, for example when the group has two connected components. If an arbitrary element  $g \in G$  cannot always be written as  $\exp(X)$ , can it nevertheless be written as  $\exp(X_1)\exp(X_2)\exp(X_3) \cdots$ ?

### 3.2.3 For a Connected Lie Group, the Local Image of the Exponential Map Generates the Whole Group

**Theorem:** If  $G$  is connected, every element  $g \in G$  of  $G$  is a finite product of exponentials: there exist  $X_1, X_2, \dots, X_k \in \mathfrak{g}$  such that

$$g = \exp(X_1) \cdot \exp(X_2) \cdot \dots \cdot \exp(X_k)$$

**Proof:** First, if  $W$  is any neighborhood of  $e$  in  $G$ , we show that  $\text{gen}(W)=G$ , where  $\text{gen}(W)$  is the subgroup of all finite products of elements of  $W$  and their inverses.

Set  $W_1 = W \cup W^{-1}$ , a symmetric neighborhood containing  $e$ , and let

$$W_k = W_1 \dots W_1$$

the product of  $k$  copies of  $W_1$ . Define

$$\text{gen}(W) = \bigcup_{k=1}^{\infty} W_k$$

By definition,  $\text{gen}(W)$  is a subgroup of  $G$ . We show that  $\text{gen}(W)$  is both open and closed.

(a)  $\text{gen}(W)$  is open.

Since  $W$  is open and inversion is a smooth bijection, hence a diffeomorphism,  $W^{-1}$  is open and so is  $W_1$ . The set  $W_k$  is a union, indexed by  $W_1$ , of images of  $W_{k-1}$  under left translations  $L_g$ :

$$W_k = \{g \cdot h \mid g \in W_1, h \in W_{k-1}\}$$

For fixed  $g \in W_1$ , the set

$$\{g \cdot h \mid h \in W_{k-1}\}$$

is exactly the image of  $W_{k-1}$  under  $L_g$ , namely  $L_g(W_{k-1})$ . Thus  $W_k$  is the union of these translates:

$$W_k = \bigcup_{g \in W_1} L_g(W_{k-1})$$

Since  $L_g$  is a diffeomorphism and preserves openness, induction from the openness of  $W_1$  shows that all  $W_k$  are open. A union of open sets is open, so  $\text{gen}(W)$  is an open subgroup of  $G$ .

(b)  $\text{gen}(W)$  is closed.

Every open subgroup of a Lie group is also closed. Consider the cosets

$$h\text{gen}(W)$$

of  $\text{gen}(W)$ . Since  $\text{gen}(W)$  is open, all cosets  $h\text{gen}(W)$  are open. Its complement is the union of all cosets distinct from  $\text{gen}(W)$  itself:

$$G \setminus \text{gen}(W) = \bigcup_{h \notin \text{gen}(W)} h\text{gen}(W)$$

This complement is a union of open sets and therefore open, so  $\text{gen}(W)$  is closed.

Thus  $\text{gen}(W)$  is nonempty (it contains  $e$ ), open, and closed. Since  $G$  is connected, its only nonempty subset that is both open and closed is  $G$  itself. Hence  $\text{gen}(W) = G$ .

For connected  $G$ , apply this result to the exponential map. Since  $\exp$  is a local diffeomorphism, choose an open neighborhood  $U$  of  $0$  whose image  $\exp(U)$  is an open neighborhood of the identity. Then  $\exp(U)$  generates  $G$ , so every  $g$  is a finite product of elements  $\exp(x_i)$ .

### 3.2.4 The Adjoint Representations of a Lie Group and Its Lie Algebra

**The Adjoint Representation of a Lie Group** The adjoint representation describes how group elements act by linear rotations or transformations on their own Lie algebra through conjugation. First define the inner automorphism of the Lie group:

$$\begin{aligned} C_g : G &\rightarrow G \\ a &\longmapsto gag^{-1} \end{aligned}$$

It clearly fixes the identity. Its differential there is called the adjoint operator

$$\text{Ad}_a = (dC_a)_e = d(L_a \circ R_{a^{-1}})_e = d(L_a)_{a^{-1}} \circ d(R_{a^{-1}})_e : \mathfrak{g} \rightarrow \mathfrak{g}$$

Next define

$$\begin{aligned} \text{Ad} : G &\rightarrow GL(\mathfrak{g}) \\ a &\longmapsto \text{Ad}_a \end{aligned}$$

This is the adjoint representation of the Lie group. One can verify that it preserves group multiplication. It sends each element  $a$  to an invertible linear transformation  $\text{Ad}_a$ . By the definition of the differential, its formula is

$$\text{Ad}_a(X) = \left. \frac{d}{dt} \right|_{t=0} (a \exp(tX) a^{-1})$$

where  $X \in \mathfrak{g}$ .

**The Adjoint Representation of a Lie Algebra** Whereas the group representation  $\text{Ad}$  describes how a group acts on vectors, the Lie algebra representation  $\text{ad}$  describes how vectors act on vectors. Viewed as a map of Lie groups,  $\text{Ad}$  sends identity to identity. Define its differential there by

$$d(\text{Ad})_e =: \text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$$

It sends a Lie algebra element  $X$  to the linear transformation  $\text{ad}_X$  of  $\mathfrak{g}$  and is called a representation of the Lie algebra. By differentiation,

$$\text{ad}_X(Y) = \left. \frac{d}{ds} \right|_{s=0} \text{Ad}_{\exp(sX)}(Y)$$

The chain rule gives  $\text{ad}_{tX} = t \text{ad}_X$ .

We shall soon see that

$$\text{Ad}_{\exp(tX)} = \exp(t \text{ad}_X)$$

and, most importantly,

$$\text{ad}_X(Y) = [X, Y]$$

The Jacobi identity implies that  $\text{ad}$  is indeed a Lie algebra representation, satisfying

$$\text{ad}_{[X,Y]} = [\text{ad}_X, \text{ad}_Y]$$

Thus  $\text{ad}$  is a Lie algebra homomorphism. The right side is the operator (matrix) commutator  $\text{ad}_X \circ \text{ad}_Y - \text{ad}_Y \circ \text{ad}_X$ .

### 3.3 Basic Relations between Lie Group and Lie Algebra Homomorphisms

#### 3.3.1 The Differential at the Identity Is a Lie Algebra Homomorphism and Commutes with the Exponential Maps

##### Commutativity of a Lie Group Homomorphism, Its Differential, and the Exponential Maps

**Theorem:** If  $f$  is a Lie group homomorphism from  $G$  to  $H$ , then  $df_e$  is a Lie algebra homomorphism, preserving brackets, and  $\forall X \in \mathfrak{g}, f \circ \exp_G(X) = \exp_H \circ df_e(X)$ . Equivalently,

the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{df_e} & \mathfrak{h} \\ \exp_G \downarrow & & \downarrow \exp_H \\ G & \xrightarrow{f} & H \end{array}$$

**Proof:**

For every  $v \in \mathfrak{g}$ , the map  $t \mapsto f(\exp_G(tv))$  is a homomorphism from  $(\mathbb{R}, +)$  to  $G$ , whose tangent at the origin satisfies  $= df_e(v)$ . On the other hand,  $t \mapsto \exp_H(t df_e(v))$  is the unique maximal integral curve in  $H$  with tangent  $df_1(v)$  at 0. Their initial points and tangents agree, so uniqueness implies that the curves coincide.

Applying this conclusion to  $\text{Ad}$  and  $\text{ad}$  gives  $\text{Ad}_{\exp(tX)} = \exp(t \text{ad}_X)$ , and hence the commutative diagram:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\text{ad}} & \mathfrak{gl}(\mathfrak{g}) \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\text{Ad}} & GL(\mathfrak{g}) \end{array}$$

**Theorem:**  $\text{ad}_X(Y) = [X, Y]$ .

**Proof:**

Since  $\text{Ad}_{\exp(sX)}(Y) = d(R_{\exp(-sX)})_{\exp(sX)} \circ d(L_{\exp(sX)})_e(Y)$ , let  $\tilde{Y}$  be the left-invariant vector field generated by  $Y \in \mathfrak{g}$ , so that  $\tilde{Y}_g = d(L_g)_e(Y)$ . The formula becomes

$$\text{Ad}_{\exp(sX)}(Y) = d(R_{\exp(-sX)})_{\exp(sX)}(\tilde{Y}_{\exp(sX)})$$

Let  $\tilde{X}$  be the left-invariant field corresponding to  $X \in \mathfrak{g}$ . Its flow is right translation:

$$\phi_s(g) = R_{\exp(sX)}(g) = g \cdot \exp(sX)$$

Substitute this differential of right translation for  $d(\phi)$  in the Lie derivative formula,

$$\mathcal{L}_X Y = \left. \frac{d}{ds} \right|_{s=0} (d\phi_{-s})(Y_{\phi_s})$$

This yields  $\mathcal{L}_{\tilde{X}} \tilde{Y} = \text{ad}_X(Y)$ . The left side equals  $[X, Y]$ , so  $\text{ad}_X(Y) = [X, Y]$ .

**Theorem:**  $df_e$  is a Lie algebra homomorphism.

**Proof:** Let  $f : G \rightarrow H$  be a Lie group homomorphism with respective Lie algebras  $\mathfrak{g}, \mathfrak{h}$ . Its differential at the identity,  $df_e : \mathfrak{g} \rightarrow \mathfrak{h}$ , preserves the Lie bracket: for any  $X, Y \in \mathfrak{g}$ ,

$$df_e([X, Y]) = [df_e(X), df_e(Y)]$$

Step 1: establish a conjugation relation using the homomorphism property. Since  $f : G \rightarrow H$  is a homomorphism, for any  $g, x \in G$  we have

$$f(gxg^{-1}) = f(g)f(x)f(g)^{-1}$$

Define the inner automorphism

$$c_g(x) = gxg^{-1}$$

on  $G$  and the inner automorphism

$$C_{f(g)}(y) = f(g)yf(g)^{-1}$$

on  $H$ . The relation becomes

$$f \circ c_g = C_{f(g)} \circ f$$

Step 2: differentiate at the identity to introduce  $\text{Ad}$ . Differentiating at  $x = e$ , the chain rule and the definition  $\text{Ad}_g = d(c_g)_e$  give

$$df_e \circ \text{Ad}_g = \text{Ad}_{f(g)} \circ df_e$$

Applying this operator identity to any  $Y \in \mathfrak{g}$  yields

$$df_e(\text{Ad}_g Y) = \text{Ad}_{f(g)}(df_e Y)$$

Step 3: parameterize using the exponential map. To introduce the Lie bracket, set  $g = \exp(tX)$  with  $X \in \mathfrak{g}$ . Substitution gives

$$df_e(\text{Ad}_{\exp(tX)} Y) = \text{Ad}_{f(\exp(tX))}(df_e Y)$$

By the naturality of  $\exp$  proved above,  $f(\exp(v)) = \exp(df_e v)$ , so we may rewrite this as

$$df_e(\text{Ad}_{\exp(tX)} Y) = \text{Ad}_{\exp(df_e X)}(df_e Y)$$

Step 4: differentiate in  $t$  to introduce  $\text{ad}$ . Differentiate in  $t$  at  $t = 0$ . Use the definition

$$\text{ad}_X = \left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(tX)}$$

and the identity  $\text{ad}_X Y = [X, Y]$ . The left side is  $\left. \frac{d}{dt} \right|_{t=0} df_e(\text{Ad}_{\exp(tX)} Y) = df_e([X, Y])$ , since linearity of  $df_e$  allows it to pass through differentiation. The right side is

$$\left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(t df_e X)}(df_e Y) = [df_e X, df_e Y]$$

We conclude that

$$df_e([X, Y]) = [df_e X, df_e Y]$$

Thus the differential of a Lie group homomorphism at the identity is a Lie algebra homomorphism.

### 3.3.2 For Connected $G$ , a Lie Group Homomorphism Is Uniquely Determined by Its Lie Algebra Homomorphism

**Theorem:** For a connected Lie group  $G$ , a Lie group homomorphism is uniquely determined by its associated Lie algebra homomorphism.

**Proof:**

Suppose  $\phi_i, i = 1, 2$  are Lie group homomorphisms with the same Lie algebra homomorphism. The exponential map is a diffeomorphism on a sufficiently small open subset  $U \subseteq \mathfrak{g}$  of  $\mathfrak{g}$  containing 0, so it has an inverse  $\exp_G^{-1}$ , the logarithm map. For every  $a \in \exp_G(U)$ ,

$$\phi_i(a) = \exp_H(d(\phi_i)_e(\exp_G^{-1}(a))), \quad i = 1, 2.$$

Since their differentials at the identity agree,  $d(\phi_1)_e = d(\phi_2)_e$ , the maps  $\phi_1$  and  $\phi_2$  coincide on the entire neighborhood  $\exp_G(U)$ . By Section 3.2.2, in any connected Lie group  $G$ , any identity neighborhood  $V = \exp_G(U)$  generates the whole group  $G$  through finite products. Thus every  $g \in G$  has the form

$$g = v_1 v_2 \dots v_n, \quad v_i \in V$$

The homomorphism property now forces the result: each  $v_i$  lies in the neighborhood, so its

image  $\phi(v_i)$  is uniquely determined by  $d\phi_1$ . Hence

$$\phi(g) = \phi(v_1)\phi(v_2) \dots \phi(v_n)$$

Every factor on the right is unique, so  $\phi(g)$  is also unique. The conclusion therefore holds globally:  $\phi_1 = \phi_2$  throughout  $G$ . Thus, for connected  $G$ , a Lie group homomorphism is uniquely determined by its Lie algebra homomorphism.

### 3.3.3 Correspondence of Isomorphisms for Simply Connected Lie Groups and Lie Algebras

**Lie Group and Lie Algebra Isomorphisms** In mathematics, isomorphic objects have exactly the same structure, differing only in names or notation.

A Lie group isomorphism ( $G \cong H$ ) must be both a group isomorphism, preserving multiplication, and a diffeomorphism, preserving smooth geometric structure. A Lie algebra isomorphism ( $\mathfrak{g} \cong \mathfrak{h}$ ) is a linear bijection  $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$  preserving the bracket:

$$\phi([X, Y]_{\mathfrak{g}}) = [\phi(X), \phi(Y)]_{\mathfrak{h}}$$

The relationship between Lie groups and Lie algebras is asymmetric. A Lie group has a complex structure, whereas its Lie algebra has a simpler linear structure. Passing from the former to the latter is natural, but the converse is not automatic:

1. From groups to algebras: always unique. Every Lie group  $G$  has a unique Lie algebra  $\mathfrak{g}$ . If two groups are isomorphic ( $G \cong H$ ), then their Lie algebras are isomorphic ( $\mathfrak{g} \cong \mathfrak{h}$ ). A Lie group isomorphism  $f$  is a global diffeomorphism, so  $(df)_e$  is a linear bijection  $\mathfrak{g} \rightarrow \mathfrak{h}$ . As already proved,  $(df)_e$  preserves brackets, and therefore is a Lie algebra isomorphism.

2. From algebras to groups: not unique. Even identical Lie algebras can correspond to different groups. For example, consider  $\mathbb{R}^n$  and  $T^n = \mathbb{R}^n/(\mathbb{Z}^n)$ ,  $\pi : \mathbb{R}^n \rightarrow T^n$ , a covering homomorphism. It is a local diffeomorphism at the origin, so  $d(\pi)_e$  is a Lie algebra isomorphism. Nevertheless,  $\pi$  is not a Lie group isomorphism because the spaces are not homeomorphic.

Under what conditions does a Lie algebra isomorphism imply a Lie group isomorphism? The answer is **when the Lie groups are simply connected**.

**Theorem:** For simply connected Lie groups, Lie group isomorphisms and Lie algebra iso-

morphisms correspond one-to-one.

**Proof:** First we must understand how smooth right and left group actions generate Lie algebra actions. Suppose the Lie group  $G$  acts smoothly on the right of a smooth manifold  $M$ , with action

$$\theta: M \times G \rightarrow M$$

$(p, g) \mapsto p \cdot g$ . Every element  $X \in \text{Lie}(G)$  determines a smooth global flow on  $M$ :

$$(t, p) \mapsto p \cdot \exp tX.$$

This is a flow because

$$\exp(0X) = e, (\exp sX)(\exp tX) = \exp(s + t)X$$

Let  $\hat{X} \in \mathfrak{X}(M)$  be its infinitesimal generator; then for every  $p \in M$ ,

$$\hat{X}_p = \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp tX.$$

We thereby obtain the map

$$\hat{\theta}: \text{Lie}(G) \rightarrow \mathfrak{X}(M)$$

defined by  $\hat{\theta}(X) = \hat{X}$ .

Another useful description of  $\hat{X}$  uses the orbit map  $\theta^{(p)}: G \rightarrow M$  defined by

$$\theta^{(p)}(g) = p \cdot g$$

Since  $\gamma(t) = \exp tX$  is a smooth curve in  $G$  with initial velocity  $\gamma'(0) = X_e$ , for every  $p \in M$  we have

$$d(\theta^{(p)})_e(X_e) = (\theta^{(p)} \circ \gamma)'(0) = \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp tX = \hat{X}_p.$$

### **Lemma: Fundamental Theorem of Lie Algebra Actions**

Let  $M$  be a smooth manifold,  $G$  a simply connected Lie group, and  $\mathfrak{g} = \text{Lie}(G)$ . Suppose  $\hat{\theta}: \mathfrak{g} \rightarrow \mathfrak{X}(M)$  is a complete homomorphism on  $M$ . Then there is a unique right  $G$ -action on  $M$  with infinitesimal generator  $\hat{\theta}$ .

Similarly, if  $G$  is simply connected, every complete antihomomorphism  $\hat{\theta}: \text{Lie}(G) \rightarrow \mathfrak{X}(M)$

is the infinitesimal generator  $\hat{\theta}$  of a unique left  $G$ -action on  $M$ .

We now prove that simply connected Lie groups with isomorphic Lie algebras are isomorphic. A Lie algebra homomorphism

$$\varphi: \mathfrak{g} \rightarrow \mathfrak{h} \subseteq \mathfrak{X}(H)$$

can be viewed as a complete  $\mathfrak{g}$ -action on  $H$ , because every left-invariant field is complete. The lemma therefore gives a unique right  $G$ -action

$$\theta: H \times G \rightarrow H$$

with infinitesimal generator  $\varphi$ . For  $X \in \mathfrak{g}$  write  $\hat{X} = \varphi(X)$ ; for  $h \in H$  and  $g \in G$  write  $h \cdot g = \theta(h, g)$ . Define the smooth map

$$\Phi: G \rightarrow H$$

by

$$\Phi(g) = e \cdot g$$

where  $e$  is the identity of  $H$ . We shall show that  $\Phi$  is the required homomorphism.

For every  $h \in H$  and  $X \in \mathfrak{g}$ , the fields  $X$  and  $\hat{X}$  are  $\theta^{(h)}$ -related. The map  $\theta^{(h)}$  takes integral curves of  $X$  to those of  $\hat{X}$ . Thus

$$t \mapsto h \cdot \exp tX$$

is the integral curve of  $\hat{X}$  starting at  $h$ .

On the other hand, since  $\hat{X}$  is left-invariant on  $H$ , left translations in  $H$  map integral curves of  $\hat{X}$  to integral curves of  $\hat{X}$ . Hence, for any  $h \in H$  and  $X \in \mathfrak{g}$ ,

$$L_h(e \cdot \exp tX) = h \cdot \exp tX$$

.

Setting  $h = e \cdot g$  for some  $g \in G$  gives:

$$(e \cdot g)(e \cdot \exp tX) = (e \cdot g) \cdot \exp tX = e \cdot g \exp tX.$$

The last equality follows because  $\theta$  is an action. Rewriting in terms of  $\Phi$  yields:

$$\Phi(g)\Phi(\exp tX) = \Phi(g \exp tX).$$

Since  $G$  is connected,  $G$  is generated by the image of  $\exp$ , so this identity implies that  $\Phi$  is a homomorphism.

To prove  $\Phi_* = \varphi$ , take arbitrary  $X \in \mathfrak{g}$ . Since  $\varphi$  is the infinitesimal generator of  $\theta$ ,

$$\varphi(X)|_e = \left. \frac{d}{dt} \right|_{t=0} (e \cdot \exp tX) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\exp tX) = d\Phi_e(X_e).$$

As  $\Phi_*$  is uniquely determined by the action of  $d\Phi_e$ , it follows that  $\Phi_*X = \varphi(X)$ . This completes the proof; the preceding result shows that  $\Phi$  is the unique homomorphism with this property.

Consequently, if  $G$  and  $H$  are simply connected Lie groups with isomorphic Lie algebras, they are isomorphic. In the earlier example  $\mathbb{R}^n$  and  $T^n = \mathbb{R}^n/(\mathbb{Z}^n)$ , the space  $T^n$  is not simply connected, so the theorem does not apply.

### 3.4 The Baker-Campbell-Hausdorff Formula: Expressing Local Group Multiplication through the Lie Bracket

The BCH formula converts multiplication of exponentials into addition and bracket operations in the Lie algebra. Theoretically, it ensures that the local structure of a Lie group is completely determined by its Lie algebra; in engineering, it provides a linearization tool for nonlinear optimization. Given two Lie algebra elements  $A$  and  $B$ , their group elements have product  $\exp(A)\exp(B)$ . Can this product be written as  $\exp(C)$ , and if so, what is  $C$ ? The BCH formula is more than an algebraic identity: it reveals how the Lie algebra, as the tangent space at the identity, locally determines group multiplication through the Lie bracket.

For  $C = \ln(\exp(A)\exp(B))$ , the BCH formula is

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] - [B, [A, B]]) + \dots$$

The exponential map is a diffeomorphism only locally near the identity, and its logarithm is likewise local. The BCH formula therefore has important applicability conditions. Specifically,  $\exp$  is a local diffeomorphism on a neighborhood  $U$  of  $\mathbf{0} \in \mathfrak{g}$ . The formula is valid only when  $A, B$  and their combination  $C$  all lie in  $U$  [3].

## 4. Matrix Lie Groups and SU(2)

An  $n \times n$  real matrix consists of  $n^2$  real numbers. Thus the space  $M_n(\mathbb{R})$  of all such matrices is topologically equivalent to  $\mathbb{R}^{n^2}$ . For complex matrices,  $M_n(\mathbb{C})$  is equivalent to  $\mathbb{C}^{n^2}$ , or  $\mathbb{R}^{2n^2}$  as a real manifold. Matrix Lie groups are often defined by constraint equations and inherit the subspace topology. The examples below show that the simple ambient structure of  $M_n(\mathbb{R}(\mathbb{C}))$  can simplify geometric constructions and make objects explicit. For example, a tangent vector (derivation) at each point can be represented explicitly as a matrix.

### 4.1 Computational Advantages of Matrix Lie Groups

**The Exponential Map and Lie Algebra of a Matrix Lie Group** Since all group elements are matrices, these definitions are intuitive and readily computable:

$$\exp(X) = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{1}{2!}X^2 + \frac{1}{3!}X^3 + \dots$$

The series converges and its result is again a matrix [3].

Let  $G$  be a matrix Lie group. Its Lie algebra  $\mathfrak{g}$  consists of all matrices  $X$  such that, for every real  $t \in \mathbb{R}$ , the matrix exponential  $e^{tX}$  lies in  $G$ . Thus

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) \mid \forall t \in \mathbb{R}, e^{tX} \in G\}$$

This definition agrees with the general definitions of the Lie algebra and exponential map. Moreover, for matrix Lie groups, the Lie bracket is the matrix commutator  $[X, Y] = XY - YX$ .

**The Differential of Left Translation in a Matrix Lie Group** To compute the pushforward of a vector or vector field under left translation, we need the differential of a map between manifolds. In a general Lie group, a group element  $g$  cannot simply be multiplied by a vector; instead, the differential of left translation  $L_g$  must act on the Lie algebra element. For matrix groups, the ambient coordinates are  $\mathbb{R}^{n^2}$  or  $\mathbb{C}^{n^2}$ , so  $L_g$  can be regarded as a linear transformation of  $\mathbb{R}^{n^2}$  or  $\mathbb{C}^{n^2}$ . For a linear function  $f(x) = Ax$ , its Jacobian  $Df$  is  $A$  itself. Hence the action of  $L_g$  on a tangent vector is multiplication of the corresponding matrix by  $g$ .

**Adjoint Representations of Matrix Lie Groups and Lie Algebras** For fixed  $g \in G$ , conjugation  $C_g : G \rightarrow G$  is defined by

$$C_g(h) = ghg^{-1} \quad (\forall h \in G)$$

Thus

$$Ad_a = (dC_a)_e = d(L_a \circ R_{a^{-1}})_e = d(L_a)_{a^{-1}} \circ d(R_{a^{-1}})_e : \mathfrak{g} \rightarrow \mathfrak{g}$$

$$X \longmapsto aXa^{-1}$$

giving an explicit expression for  $Ad_a$ .

Differentiating  $Ad$  at the identity gives the Lie algebra adjoint representation  $ad$ , satisfying  $ad_Y(X) = YX - XY = [Y, X]$ .

**Theorem 3.3.1 for Matrix Lie Groups** Theorem 3.3.1 takes a particularly simple form in matrix notation. Let  $G$  and  $H$  be matrix Lie groups with respective Lie algebras  $\mathfrak{g}$  and  $\mathfrak{h}$ , and let  $\Phi : G \rightarrow H$  be a Lie group homomorphism. There is a unique real-linear map  $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$  such that, for all  $X \in \mathfrak{g}$ , the following hold:

1.  $\Phi(e^X) = e^{\phi(X)}$
2.  $\phi(AXA^{-1}) = \Phi(A)\phi(X)\Phi(A)^{-1}$  for all  $X \in \mathfrak{g}, A \in G$ .
3.  $\phi([X, Y]) = [\phi(X), \phi(Y)]$  for all  $X, Y \in \mathfrak{g}$ .
4.  $\phi(X) = \left. \frac{d}{dt} \Phi(e^{tX}) \right|_{t=0}$  for all  $X \in \mathfrak{g}$ .

## 4.2 An Example: SU(2)

Merely stating theorems is insufficient for deeper understanding. We therefore examine  $SU(2)$ , a classical Lie group in physics.

$$SU(2) = \{U \in M_{2 \times 2}(\mathbb{C}) \mid U^\dagger U = I, \det(U) = 1\}$$

Its elements have the form  $U = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix}$ , where  $\alpha, \beta \in \mathbb{C}$  and the normalization condition  $|\alpha|^2 + |\beta|^2 = 1$  holds. As a subset of  $M_2(\mathbb{C})$ , it lies in  $C^4$ , or  $R^8$ , and its topology should therefore be inherited from  $R^8$ . However, the constraints show that the entries are determined by only two complex numbers, or four real numbers, leaving redundant dimensions.

To avoid this redundancy, define  $f : \mathbb{C}^2 \rightarrow \mathbb{C}^4$  by mapping two complex numbers to the four entries of a  $2 \times 2$  matrix:

$$f(\alpha, \beta) = (\alpha, \beta, -\bar{\beta}, \bar{\alpha})$$

This preserves Euclidean distance. A basic topological lemma states that if  $X \subset Y \subset Z$ , the subspace topology on  $X$  inherited from  $Y$  agrees with that on  $X$  inherited directly from  $Z$ . Thus  $SU(2)$  can be regarded directly as a subspace of  $\mathbb{C}^2$ , with the subspace topology inherited from  $\mathbb{C}^2$ . Equivalently,

$$SU(2) := \{(\alpha, \beta) \in \mathbb{C}^2, |\alpha|^2 + |\beta|^2 = 1\}$$

which makes the topology inherited from  $\mathbb{C}^2$  clear. Writing

$$\alpha = a + bi, \beta = c + di$$

the normalization condition becomes

$$a^2 + b^2 + c^2 + d^2 = 1$$

Define  $g : \mathbb{C}^2 \rightarrow \mathbb{R}^4$  by

$$g\left(\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}\right) = (\operatorname{Re}(z_1), \operatorname{Im}(z_1), \operatorname{Re}(z_2), \operatorname{Im}(z_2))$$

Then  $g : SU(2) \rightarrow S^3$  is continuous and bijective, so  $SU(2)$  is naturally homeomorphic to  $S^3$ . Hence it is connected and, moreover, simply connected. Since Chapter 1 showed that  $SO(3)$  is  $RP^3$ ,  $SU(2)$  is the two-sheeted universal cover of  $RP^3$ .

Using stereographic projection, we can write the coordinate parametrizations  $\varphi_N : \mathbb{R}^3 \rightarrow SU(2)$  and  $\varphi_S : \mathbb{R}^3 \rightarrow SU(2)$  for  $SU(2)$ :

$$\varphi_N(x, y, z) = \begin{pmatrix} \frac{2x}{x^2+y^2+z^2+1} + \frac{2yi}{x^2+y^2+z^2+1} & \frac{2z}{x^2+y^2+z^2+1} + \frac{i(x^2+y^2+z^2-1)}{x^2+y^2+z^2+1} \\ -\frac{2z}{x^2+y^2+z^2+1} + \frac{i(x^2+y^2+z^2-1)}{x^2+y^2+z^2+1} & \frac{2x}{x^2+y^2+z^2+1} - \frac{2yi}{x^2+y^2+z^2+1} \end{pmatrix}$$

and

$$\varphi_S(x, y, z) = \begin{pmatrix} \frac{2x}{x^2+y^2+z^2+1} + \frac{2yi}{x^2+y^2+z^2+1} & \frac{2z}{x^2+y^2+z^2+1} + \frac{i(1-(x^2+y^2+z^2))}{x^2+y^2+z^2+1} \\ -\frac{2z}{x^2+y^2+z^2+1} + \frac{i(1-(x^2+y^2+z^2))}{x^2+y^2+z^2+1} & \frac{2x}{x^2+y^2+z^2+1} - \frac{2yi}{x^2+y^2+z^2+1} \end{pmatrix}$$

**The Lie Algebra of  $SU(2)$**  By the method above for matrix Lie groups, the Lie algebra  $\mathfrak{su}(2)$  is the tangent space of  $SU(2)$  at the identity  $I$ : the set of matrices  $X$  for which a smooth curve

$$U(t) \in SU(2), U(0) = I, U'(0) = X$$

exists. Take a curve  $U(t) \in SU(2)$  with  $U(0) = I$ . Differentiate  $U(t)^\dagger U(t) = I$  in  $t$  at  $t = 0$ , using the commutation of differentiation with conjugate transpose, to obtain

$$\left. \frac{d}{dt} \right|_{t=0} (U(t)^\dagger U(t)) = X^\dagger I + I X = X^\dagger + X = 0 \implies X^\dagger = -X.$$

Differentiating  $\det U(t) = 1$  in  $t$  and applying Jacobi's formula at  $t = 0$  gives

$$\left. \frac{d}{dt} \right|_{t=0} \det U(t) = \det I \cdot \operatorname{tr} (I^{-1} X) = \operatorname{tr} X = 0.$$

Therefore

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) \mid X^\dagger = -X, \operatorname{tr} X = 0\}.$$

Such matrices are called skew-Hermitian: their conjugate transpose is their negative. Explicitly,

$$X = \begin{pmatrix} i\alpha & \beta + i\gamma \\ -\beta + i\gamma & -i\alpha \end{pmatrix}, \quad \alpha, \beta, \gamma \in \mathbb{R}.$$

Introduce the Pauli matrices  $\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ , and  $\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ . Then

$$X = \alpha i \sigma_3 + \beta i \sigma_2 + \gamma i \sigma_1$$

In physics one writes  $X = -i\frac{\theta}{2} \left[ n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right]$ , assigning the physical interpretation of rotation about an axis. The exponential map is  $\exp : \mathfrak{su}(2) \rightarrow SU(2)$

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

where  $X$  is skew-Hermitian. For  $X = -i\frac{\theta}{2}(\mathbf{\hat{n}} \cdot \boldsymbol{\sigma})$ , its exponential is

$$\exp \left( -i\frac{\theta}{2} \mathbf{\hat{n}} \cdot \boldsymbol{\sigma} \right) = \cos \left( \frac{\theta}{2} \right) I - i \sin \left( \frac{\theta}{2} \right) (\mathbf{\hat{n}} \cdot \boldsymbol{\sigma})$$

In expanded form this is

$$\exp(X) = \begin{pmatrix} \cos \frac{\theta}{2} - in_z \sin \frac{\theta}{2} & -(n_y + in_x) \sin \frac{\theta}{2} \\ (n_y - in_x) \sin \frac{\theta}{2} & \cos \frac{\theta}{2} + in_z \sin \frac{\theta}{2} \end{pmatrix}$$

Clearly, the right-hand side is skew-Hermitian.

Each element of  $\mathfrak{su}(2)$  corresponds to a one-parameter subgroup  $\gamma_X(t) = \exp(tX)$ . The one-parameter subgroups associated with the basis  $E_j = -\frac{i}{2}\sigma_j$  are:

1.  $X = -\frac{i}{2}\sigma_1, \gamma_1(t) = \exp(-i\frac{t}{2}\sigma_1) = \begin{pmatrix} \cos \frac{t}{2} & -i \sin \frac{t}{2} \\ -i \sin \frac{t}{2} & \cos \frac{t}{2} \end{pmatrix}$
2.  $X = -\frac{i}{2}\sigma_2, \gamma_2(t) = \exp(-i\frac{t}{2}\sigma_2) = \begin{pmatrix} \cos \frac{t}{2} & -\sin \frac{t}{2} \\ \sin \frac{t}{2} & \cos \frac{t}{2} \end{pmatrix}$
3.  $X = -\frac{i}{2}\sigma_3, \gamma_3(t) = \exp(-i\frac{t}{2}\sigma_3) = \begin{pmatrix} e^{-it/2} & 0 \\ 0 & e^{it/2} \end{pmatrix}$

**Left-Invariant Vector Fields on  $SU(2)$**  As discussed earlier, each left-invariant field corresponds to a Lie algebra element extended by left translation. Every left-invariant field  $X$  is uniquely determined by a generator in  $\mathfrak{su}(2)$ . Choose

$$E_3 = -\frac{i}{2}\sigma_3 = \begin{pmatrix} -i/2 & 0 \\ 0 & i/2 \end{pmatrix}$$

The induced field  $\tilde{E}_3$  associated with  $E_3$  has value at  $g$  given by left translation:

$$\tilde{E}_3(g) = g \cdot E_3 = \begin{pmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{pmatrix} \begin{pmatrix} -i/2 & 0 \\ 0 & i/2 \end{pmatrix} = \frac{i}{2} \begin{pmatrix} -\alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix}$$

Since  $SU(2)$  is homeomorphic to  $S^3$ , we can express both its elements and its vector fields in the coordinates and tangent bundle of  $R^4$ . Again construct a map  $\Phi$ . Associate a vector  $\vec{x} = (x_1, x_2, x_3, x_4)$  in  $\mathbb{R}^4$  with a matrix  $U$  in  $SU(2)$  via  $\Phi$ :

$$\Phi(x_1, x_2, x_3, x_4) = \begin{pmatrix} x_1 + ix_2 & x_3 + ix_4 \\ -x_3 + ix_4 & x_1 - ix_2 \end{pmatrix}$$

Under this map, elements of  $SU(2)$  correspond exactly to unit vectors in  $\mathbb{R}^4$ , namely  $S^3$ . The correspondence from  $\mathfrak{su}(2)$  to tangent vectors in  $\mathbb{R}^4$  is given by the pushforward at the identity. Choose a basis element of  $\mathfrak{su}(2)$ , for example  $X = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ , and construct a curve through  $I$ :

$$U(t) = e^{tX} = \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix}$$

equivalently

$$U(t) = \begin{pmatrix} \cos t + i \sin t & 0 \\ 0 & \cos t - i \sin t \end{pmatrix}$$

In  $\mathbb{R}^4$  this means  $x_1(t) = \cos t$ ,  $x_2(t) = \sin t$ ,  $x_3(t) = 0$ , and  $x_4(t) = 0$ . Thus  $\gamma(t) = (\cos t, \sin t, 0, 0)$

$$\vec{v} = \gamma'(0) = \left( \frac{d}{dt} \cos t, \frac{d}{dt} \sin t, 0, 0 \right) \Big|_{t=0} = (0, 1, 0, 0)$$

Now take any point  $U = \begin{pmatrix} x_1 + ix_2 & x_3 + ix_4 \\ -x_3 + ix_4 & x_1 - ix_2 \end{pmatrix}$ . The path through  $U$  with velocity determined by  $B$  is

$$\gamma(t) = U \cdot e^{tB}$$

Differentiating at  $t = 0$  gives

$$V = \frac{d}{dt}(U \cdot e^{tB}) \Big|_{t=0} = U \cdot B$$

Direct matrix multiplication yields

$$V = \begin{pmatrix} x_1 + ix_2 & x_3 + ix_4 \\ -x_3 + ix_4 & x_1 - ix_2 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} -x_2 + ix_1 & -x_4 + ix_3 \\ -x_4 - ix_3 & x_2 + ix_1 \end{pmatrix}$$

Hence the tangent vector in  $\mathbb{R}^4$  induced by this basis element at  $\vec{x}$  is

$$\vec{v} = (-x_2, x_1, -x_4, x_3)$$

which is indeed tangent to  $S^3$ .

The corresponding left-invariant field is the differential operator on  $\mathbb{R}^4$  given by

$$X_3 = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2} - x_4 \frac{\partial}{\partial x_3} + x_3 \frac{\partial}{\partial x_4}$$

restricted to  $S^3$ . At each point

$$(x_1, x_2, x_3, x_4)$$

of the sphere it points in the direction

$$(-x_2, x_1, -x_4, x_3)$$

Moreover, the matrix commutator  $[X, Y]$  agrees with the Lie bracket of the corresponding fields on  $S^3$  embedded in Euclidean space, so the descriptions are consistent.

**The Bi-Invariant Metric on SU(2)** Since SU(2) is homeomorphic to  $S^3$ , it is compact and admits a bi-invariant metric. Choose at the identity

$$\langle X, Y \rangle = 1/2 \operatorname{tr}(X^\dagger Y)$$

where  $X$  and  $Y$  are Lie algebra elements. It suffices to verify Ad-invariance:

$$\langle \operatorname{Ad}_g X, \operatorname{Ad}_g Y \rangle = \langle X, Y \rangle$$

Here  $\operatorname{Ad}_g X = gXg^{-1}$  and  $\langle \operatorname{Ad}_g X, \operatorname{Ad}_g Y \rangle = 1/2 \operatorname{Tr}((gXg^{-1})^\dagger (gYg^{-1})) = 1/2 \operatorname{Tr}(X^\dagger Y)$ , so

$$\langle \operatorname{Ad}_g X, \operatorname{Ad}_g Y \rangle = \langle X, Y \rangle$$

Thus it does give a bi-invariant metric. When the Lie algebra or vector fields are regarded as real tangent vectors to  $S^3$ , this metric is exactly the standard round metric inherited from Euclidean space. We verify this below:

$$X = \begin{pmatrix} i\alpha_1 & \beta_1 + i\gamma_1 \\ -\beta_1 + i\gamma_1 & -i\alpha_1 \end{pmatrix}, \quad \alpha_1, \beta_1, \gamma_1 \in \mathbb{R}.$$

$$Y = \begin{pmatrix} i\alpha_2 & \beta_2 + i\gamma_2 \\ -\beta_2 + i\gamma_2 & -i\alpha_2 \end{pmatrix}, \quad \alpha_2, \beta_2, \gamma_2 \in \mathbb{R}.$$

These are elements of  $\mathfrak{su}(2)$ . Their inner product  $\langle X, Y \rangle = 1/2 \operatorname{tr}(X^\dagger Y)$  is

$$\frac{1}{2} \operatorname{tr} \begin{pmatrix} \alpha_1 \alpha_2 + \beta_1 \beta_2 - i\gamma_2 \beta_1 + i\gamma_1 \beta_2 + \gamma_1 \gamma_2 & * \\ * & \alpha_1 \alpha_2 + \beta_1 \beta_2 + i\gamma_2 \beta_1 - i\gamma_1 \beta_2 + \gamma_1 \gamma_2 \end{pmatrix}$$

This equals

$$(\alpha_1 \alpha_2 + \beta_1 \beta_2 + \gamma_1 \gamma_2).$$

The vector in  $\mathbb{R}^4$  corresponding to  $X$  is the tangent vector  $(0, \alpha_1, \beta_1, \gamma_1)$  at  $(1, 0, 0, 0)$  on the sphere, while  $Y$  corresponds to  $(0, \alpha_2, \beta_2, \gamma_2)$ . Their standard Euclidean inner product is exactly  $(\alpha_1 \alpha_2 + \beta_1 \beta_2 + \gamma_1 \gamma_2)$ , showing that the definition is natural.

By the earlier result on the Levi-Civita connection of compact Lie groups, left-invariant fields satisfy

$$\nabla_X Y = \frac{1}{2} [X, Y]$$

We now compute the sectional curvature of  $SU(2)$ . Choose

$$e_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

$$\langle e_1, e_1 \rangle = -\frac{1}{2} \text{tr}(e_1^2) = -\frac{1}{2} \text{tr} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\frac{1}{2}(-2) = 1$$

Similarly,  $\langle e_2, e_2 \rangle = 1$ ,  $\langle e_3, e_3 \rangle = 1$ , and distinct basis elements have zero inner product, namely  $\langle e_i, e_j \rangle = \delta_{ij}$ . Direct computation gives the commutators

$$[e_1, e_2] = 2e_3, \quad [e_2, e_3] = 2e_1, \quad [e_3, e_1] = 2e_2$$

$$\|[e_1, e_2]\|^2 = \|2e_3\|^2 = 4\|e_3\|^2 = 4$$

For a Lie group with a bi-invariant metric, the sectional curvature of a plane spanned by orthonormal vectors  $X, Y$  is simply

$$K(X, Y) = \frac{1}{4} \|[X, Y]\|^2$$

Taking  $e_1, e_2$  as representatives (any two of the three may be chosen) gives

$$K(e_1, e_2) = \frac{1}{4} \|[e_1, e_2]\|^2$$

$$K = \frac{1}{4} \times 4 = 1$$

Since the metric is bi-invariant, the norm of the bracket is preserved under adjoint transformations. Thus every tangent plane at every point has

$$K \equiv 1$$

agreeing with the constant curvature of  $S^3$ .

Here geodesics are given by the exponential map, and both are complete. We can write an explicit geodesic with initial velocity

$$X = -\frac{i}{2}\sigma_1, \gamma_1(t) = \exp\left(-i\frac{t}{2}\sigma_1\right) = \text{Exp}\left(-i\frac{t}{2}\sigma_1\right) = \begin{pmatrix} \cos \frac{t}{2} & -i \sin \frac{t}{2} \\ -i \sin \frac{t}{2} & \cos \frac{t}{2} \end{pmatrix}$$

**Verifying the BCH Formula with  $SU(2)$**  We now use  $SU(2)$  to test the accuracy of the BCH formula.

Choose two elements of the Lie algebra of  $SU(2)$ :

$$X = \varepsilon i \sigma_1, \quad Y = \varepsilon i \sigma_2$$

First use exact matrix multiplication and the logarithm map.

The exponential formula for  $SU(2)$  is

$$\exp(i\theta \vec{n} \cdot \vec{\sigma}) = \cos \theta I + i \sin \theta (\vec{n} \cdot \vec{\sigma})$$

Set

$$\begin{aligned} M &= e^X e^Y = (\cos \varepsilon I + i \sin \varepsilon \sigma_1)(\cos \varepsilon I + i \sin \varepsilon \sigma_2) \\ &= (\cos^2 \varepsilon) I + i(\sin \varepsilon \cos \varepsilon) \sigma_1 + i(\sin \varepsilon \cos \varepsilon) \sigma_2 - (\sin^2 \varepsilon) \sigma_3 \end{aligned}$$

Suppose  $M$  is the exponential of a vector  $Z$ , so

$$M = \cos \theta I + i \sin \theta (\vec{n} \cdot \vec{\sigma})$$

Comparing components gives:

1.  $\cos \theta = \cos^2 \varepsilon$
2.  $\sin \theta n_1 = \sin \varepsilon \cos \varepsilon$
3.  $\sin \theta n_2 = \sin \varepsilon \cos \varepsilon$
4.  $\sin \theta n_3 = -\sin^2 \varepsilon$

As  $\varepsilon \rightarrow 0$ , Taylor expansion gives the components of  $\log M$ :

1.  $\theta \approx \sqrt{2} \varepsilon \sqrt{1 - \frac{\varepsilon^2}{4}} \approx \sqrt{2} \varepsilon (1 - \frac{\varepsilon^2}{8}) = \sqrt{2} \varepsilon - \frac{\sqrt{2}}{8} \varepsilon^3$
2.  $\theta n_1 \approx \varepsilon - \frac{\varepsilon^3}{3}$
3.  $\theta n_2 \approx \varepsilon - \frac{\varepsilon^3}{3}$
4.  $\theta n_3 \approx -\varepsilon^2$

Hence

$$\log(e^X e^Y) \approx i \left[ \left( \varepsilon - \frac{\varepsilon^3}{3} \right) \sigma_1 + \left( \varepsilon - \frac{\varepsilon^3}{3} \right) \sigma_2 - \varepsilon^2 \sigma_3 \right]$$

or equivalently  $Z \approx i \left[ \left( \varepsilon - \frac{\varepsilon^3}{3} \right) \sigma_1 + \left( \varepsilon - \frac{\varepsilon^3}{3} \right) \sigma_2 - \varepsilon^2 \sigma_3 \right]$ .

Alternatively, apply the BCH formula directly:  $Z = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] + \frac{1}{12}[Y, [Y, X]] + \dots$ , where

$$X + Y = i\varepsilon\sigma_1 + i\varepsilon\sigma_2, [X, Y] = [\varepsilon i\sigma_1, \varepsilon i\sigma_2] = -\varepsilon^2[\sigma_1, \sigma_2] = -2i\varepsilon^2\sigma_3$$

$[X, [X, Y]] = -4i\varepsilon^3\sigma_2$ , and  $[Y, [Y, X]] = -4i\varepsilon^3\sigma_1$ . The result

$$Z \approx i\varepsilon(\sigma_1 + \sigma_2) - i\varepsilon^2\sigma_3 - \frac{i\varepsilon^3}{3}(\sigma_1 + \sigma_2)$$

agrees with the preceding calculation and illustrates the local validity of the BCH formula.

## 5. Conclusion

Compared with general manifolds, Lie groups possess the additional rigidity and symmetry of a group structure, which give rise to favorable properties. Left translations make Lie groups far more homogeneous than general manifolds. The triviality of their tangent bundles allows complicated geometric quantities to be extended and simplified by algebraic operations. Through the exponential map and the adjoint representation, we have seen how linear tools become a key to studying nonlinear manifolds. The correspondence between Lie groups and Lie algebras offers an algebraic route to investigating the topological properties of groups. Matrix Lie groups, a prominent topic in physics and mathematics, permit simpler calculations and more explicit representations than general Lie groups. In mathematics, Lie groups closely unite algebra, geometry, and analysis, giving them exceptional significance across disciplines. In physics and engineering, matrix Lie groups likewise occupy an indispensable position.

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